

Kiyoung Om

Ph.D. Student, Electrical and Computer Engineering, The University of Texas at Austin

Machine Intelligence through Decision-making and Interaction (MIDI) Lab | Advisor: Prof. Amy Zhang

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RESEARCH INTERESTS

- Generative Models for Optimization and Control
- Alignment of Generative Models via Search and Fine-tuning

EDUCATION

The University of Texas at Austin | Aug 2026 - Present

Ph.D. in Electrical and Computer Engineering | Austin, TX, USA

Advisor: Prof. Amy Zhang, MIDI Lab

Korea Advanced Institute of Science and Technology (KAIST) | Mar 2024 - Feb 2026

M.S. in Data Science | Daejeon, South Korea | GPA: 3.78/4.3

Graduate School of Data Science | Advisor: Prof. Jinkyoo Park, System Intelligence Lab

Korea University | Mar 2018 - Feb 2024

B.S. in Industrial Management Engineering | Seoul, South Korea

GPA: 4.02/4.5 | Major GPA: 4.19/4.5

Military Service: Republic of Korea Army, Sep 2019 - Feb 2021

PUBLICATIONS

* Equal contribution.

Conference Papers & Workshop Proceedings

Diffusion Fine-Tuning via Reparameterized Policy Gradient of the Soft Q-Function

Hyeongyu Kang*, Jaewoo Lee*, Woocheol Shin*, **Kiyoung Om**, and Jinkyoo Park.

ICLR (Poster), 2026. [Paper](#) | [arXiv](#) | [Code](#)

Diffusion Alignment as Variational Expectation-Maximization

Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, Woocheol Shin, Taeyoung Yun, **Kiyoung Om**, and Jinkyoo Park.

ICLR (Poster), 2026. [Paper](#) | [arXiv](#) | [Code](#)

Posterior Inference with Diffusion Models for High-dimensional Black-box Optimization

Taeyoung Yun*, **Kiyoung Om***, Jaewoo Lee, Sujin Yun, and Jinkyoo Park.

ICML (Poster); FPI @ ICLR Workshop, 2025. [Paper](#) | [Code](#)

Posterior Inference in Latent Space for Scalable Constrained Black-box Optimization

Kiyoung Om*, Kyuil Sim*, Taeyoung Yun*, Hyeongyu Kang, and Jinkyoo Park.

NeurIPS SPIGM Workshop (Oral), 2025. [Paper](#) | [arXiv](#) | [Code](#)

Preprints

Flow-ERD: Agent-type Aware Flow Matching with Entropy-Regularized Distillation for Diverse Traffic Simulation

Seulbin Hwang*, **Kiyoung Om***, Daejung Kim, and Jinhan Lee.

arXiv preprint, 2026. [Paper](#) | [Project](#)

Manuscripts & Industry White Papers

Jung, J., Sim, Y., Lee, J. H., **Om, K.**, Shin, H., & Kwon, C. (2025). *AI-accelerated Heuristics for Production Planning*. Samsung Electronics-KAIST industry white paper. Under review.

EXPERIENCE

NAVER LABS, Robotics & Embodied AI Team | Mar 2026 - Jul 2026

Research Intern | Seongnam, South Korea

Researched learning-based traffic simulation for autonomous driving, focusing on closed-loop covariate shift and realistic multi-agent behavior.

Contributed to Flow-ERD, a flow matching based traffic simulator that ranked 1st on the [WOSAC 2025 Sim Agents Challenge leaderboard](#) (July 2026).

PROJECTS

Samsung Electronics-KAIST AI Industry-Academia Collaboration | Mar 2025 - Sep 2025

AI Research Assistant | Daejeon, South Korea

Developed an AI-accelerated Lagrangian heuristic for single-site CLSP with production leveling and joint capacity constraints.

Predicted near-optimal Lagrangian multipliers to warm-start subgradient updates; achieved approximately 75% runtime reduction vs. primal MILP, a gap below 0.1%, and approximately 96% plan similarity to optimal.

Korea University Solar Cell Lab AI Industry-Academia Collaboration | Apr 2023 - Jul 2023

AI Application Advisor | Seoul, South Korea

Developed CNN-based prediction models for weighted-average reflection of solar cells using panel images.

Created a system to predict experimental dates and efficiency for perovskite solar cells based on weather forecasts.

TEACHING EXPERIENCE

KAIST, Industrial & Systems Engineering | Mar 2024 - Jun 2024

Teaching Assistant, IE343 Statistical Machine Learning | Daejeon, South Korea

Supported undergraduate students in developing a foundation in machine learning; graded assignments and provided detailed feedback.

Covered supervised and unsupervised learning, regression, classification, and model evaluation.

Korea University | Jul 2023 - Aug 2023

Teaching Assistant, Deep Learning Programming Study | Seoul, South Korea

Assisted students with implementing neural networks using PyTorch under Prof. Sungbin Lim; guided practical exercises in deep learning.

AWARDS

Academic Excellence Award | Korea University, 2022

Capstone Excellence Award | Korea University, 2022

Topic: Understanding the Lifecycle and Gap Technologies of Lithium-Ion Batteries

Commander's Commendation | Republic of Korea Army, 2020

Speech Contest on Practicing the Core Values of the Army and Showcasing Examples

EXTRACURRICULAR ACTIVITIES

KUBIG (Korea University BIG data research club) | Jul 2022 - Jul 2023

Regular Member | Seoul, South Korea

Participated in and led study groups on NLP, computer vision, and reinforcement learning.

Developed a vehicle hazard detection system using YOLOv7.

Led an algorithmic trading and backtesting research project using FinRL.